

DD: EM-02 — Contractor & Staff Registry

Developer implementation guide: DD_EM-02-Contractor_and_Staff_Registry-DEV_IMPL_GUIDE-v1.0.md.

Canonical note (2026-06-01): this is the canonical EM-02 implementation target. It supersedes DD_EM-02-Contractor_Team_Agent_Registry-v1.0.md, which is retained only for historical traceability. Build one EM-02 capability from this contractor/staff/capacity model.

1. Purpose

This rewritten DD is the developer-facing version of the module contract DD_EM-02-Contractor_and_Staff_Registry-v1.0.md.

Verdict: ■ New | **Wave:** 2 | **Group E:** Employment Management | **Version:** v1.0 | **Module:** Contractor & Staff Registry The system of record for **who can do field work, where, and when:** contractor companies + their staff, the geographic regions and service scopes each contractor covers, the technical skills each contractor is qualified for, the time-window availability slots each contractor accepts work in with concurrency caps, the per-WO slot commitments that consume that capacity, and the main handler (one per contractor) plus optional staff roster the operator uses for contractor communication. Owned at operator scope. Referenced by WO module Drools routing rules (the assign-contractor rule queries EM-02 to find an available qualified contractor for a given region + scope + datetime + required skills), by ILM-CFG-02 cross-DD only (ILM-CFG-02 provides region IDs that EM-02 references; no reverse FK), by OSR-01 (contractor_id referenced on stock_location.contractor_id for vans; EM-02 retires osr_contractor_stub), and by ICN-01 (staff foundation_auth_user_id linkage for notification dispatch). v1.0 scope: full contractor + scope + skills + availability + slot commitment + staff (main handler mandatory, optional roster) model. The slot ledger is owned by EM-02 (per user decision earlier in design discussion — one place owns both capacity and commitments). **Per-staff skill certification is explicitly out of scope** — BSS is not an HR or training system; staff qualification tracking belongs in the operator's HR / training systems.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/10_other/DD_EM-02-Contractor_and_Staff_Registry-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- DELETE /api/contractor-slot-commitments/{id}
- GET /api/contractor-slot-commitments/{id}
- GET /api/contractor-slot-commitments?slotId=X&date=Y
- GET /api/contractor-staff/{staff_id}
- GET /api/contractor-staff?contractorId=X&active=true
- GET /api/contractor-staff?contractorId=X&staffRoleTier=MAIN_HANDLER&active=true
- GET /api/contractor-staff?contractorId=X&staffRoleTier=OPTIONAL_ROSTER&active=true
- GET /api/contractors/{contractor_id}
- GET /api/contractors/{contractor_id}/availability-slots
- GET /api/contractors/{contractor_id}/main-handler
- GET /api/contractors/{contractor_id}/region-scopes
- GET /api/contractors/{contractor_id}/region-skills
- GET /api/contractors/{id}/main-handler
- GET /api/contractors?operatorCode=WIK&active=true&contractorType=FRANCHISE
- GET /api/contractors?operatorCode=X
- GET /api/skill-catalog?operatorCode=WIK&active=true
- PATCH /api/contractor-availability-slots/{slot_id}
- PATCH /api/contractor-staff/{staff_id}
- PATCH /api/contractors/{contractor_id}
- PATCH /api/skill-catalog/{operator}/{skill_code}
- POST /api/contractor-availability
- POST /api/contractor-slot-commitments
- POST /api/contractor-slot-commitments/{id}/consume
- POST /api/contractor-staff
- POST /api/contractors
- POST /api/contractors/{contractor_id}/availability-slots
- POST /api/contractors/{contractor_id}/main-handler/handover
- POST /api/contractors/{contractor_id}/region-scopes
- POST /api/contractors/{contractor_id}/region-skills
- POST /api/skill-catalog

4.2 Tables

- contractor
- contractor_availability_slot
- contractor_region_scope
- contractor_region_skill
- contractor_slot_commitment
- contractor_staff
- skill_catalog

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;

- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- assign-contractor
- contractor-availability
- dwdm-config
- fiber-install
- ftth-cpe
- huawei-academy
- iptv-stb-setup
- network-troubleshooting
- sophix-superadmin
- vehicle-license-class-b
- vip-handling
- wik-customer-service-training
- wik-franchise-onboarding
- wik-internal-tech-team
- wik-internal-training
- yassn-training

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- No domain events are explicitly declared in this DD.

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.

8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

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The system of record for **who can do field work, where, and when:** contractor companies + their staff, the geographic regions and service scopes each contractor covers, the technical skills each contractor is qualified for, the time-window availability slots each contractor accepts work in with concurrency caps, the per-WO slot commitments that consume that capacity, and the main handler (one per contractor) plus optional staff roster the operator uses for contractor communication.

Owned at operator scope. Referenced by WO module Drools routing rules (the assign-contractor rule queries EM-02 to find an available qualified contractor for a given region + scope + datetime + required skills), by ILM-CFG-02 cross-DD only (ILM-CFG-02 provides region IDs that EM-02 references; no reverse FK), by OSR-01 (contractor_id referenced on stock_location.contractor_id for vans; EM-02 retires osr_contractor_stub), and by ICN-01 (staff foundation_auth_user_id linkage for notification dispatch).

v1.0 scope: full contractor + scope + skills + availability + slot commitment + staff (main handler mandatory, optional roster) model. The slot ledger is owned by EM-02 (per user decision earlier in design discussion — one place owns both capacity and commitments). **Per-staff skill certification is explicitly out of scope** — BSS is not an HR or training system; staff qualification tracking belongs in the operator's HR / training systems.

1. What this module is

When WO module asks **"who can install fiber in KE-NRB-KAREN on Monday 2026-02-23 at 14:00 with skills fiber-install + ftth-cpe + vip-handling, and has capacity?"**, the answer is structural: there is a list of contractors covering KE-NRB-KAREN for service scope INSTALL, each with a skill profile and an availability slot schedule. EM-02 owns that structural data.

The model is intentionally **contractor-anchored — not staff-anchored**. A contractor commits to a coverage window with a concurrency cap (e.g., "Franchise A van accepts up to 5 concurrent install WOs on Mondays 09:00–18:00"); how Franchise A internally staffs that capacity (which crew, which technician, which truck) is Franchise A's problem. The operator schedules against the slot, decrements the capacity at commitment, and trusts the contractor to deliver.

The main-handler model

For staff tracking, EM-02 takes a pragmatic two-tier approach reflecting the operational reality that **external contractor staff churn is high** and trying to mirror each contractor's internal HR moves is fragile:

- **MAIN_HANDLER** (mandatory) — every active contractor MUST have exactly one active MAIN_HANDLER staff record. The main handler is the operator's spoke into that contractor: the person of contact for assignments, slot scheduling, dispatching, escalation, and acknowledgments. ICN-01 dispatches contractor-facing notifications to the main handler's FOUNDATION_AUTH user. If the main handler leaves the contractor, the contractor's BO admin updates the main handler before the change takes effect (handover event tracked).
- **OPTIONAL_ROSTER** (opt-in per contractor) — additional staff records for contractors who want full roster visibility in the operator system. Typically populated for INTERNAL contractors (where Wananchi is the employer) and for some large franchises with sophisticated ops. For most franchises, this stays empty — and that's fine. The operator system does not need to know every technician's name to schedule work.

WO routing does not depend on individual staff data. The Drools assign-contractor rule filters by `contractor_region_skill` (contractor-level certification) — the question is "does this contractor have the skill in this region", not "does this specific technician have the skill". Per-person qualification tracking is not in EM-02 scope.

What this module does

1. **Catalog contractors** — operator-scoped, with identity (legal entity name, registration number, type — FRANCHISE / INTERNAL / SUB_CONTRACTOR), contract reference, contact details, performance reference (link to EM-04 Reporting metrics, pending).
2. **Define contractor coverage** — per (contractor, tech_region_id, service_scope) row with effective dates. A contractor can be INSTALL+SUPPORT in one region but only SUPPORT in another.
3. **Maintain skill catalog** — operator-extensible technical skill codes (fiber-install, ftth-cpe, dwdm-config, vip-handling, iptv-stb-setup, etc.). Operator can add new codes via config.
4. **Track contractor skill profiles per region** — which skills a contractor is certified to perform in each region they cover. **This is what WO routing filters on.** Different regions may require different skill mixes.
5. **Define availability slots** — day-of-week + hour-range windows with concurrency caps, per (contractor, region, service_scope). Multiple slots per contractor allowed (Monday morning + Monday afternoon as separate rows with different caps).
6. **Maintain slot commitment ledger** — immutable record of every WO committed against a slot, with consumption + release. The slot's "remaining capacity at time T" is derived from the ledger.
7. **Resolve availability** — POST /api/contractor-availability is the WO module's hot path: given (region, scope, required skills, datetime window, demand qty), return ranked list of contractors with capacity, with slot identifiers to consume.
8. **Track contractor main handlers** — exactly one main handler per active contractor, mandatory; the spoke for all operator-contractor communication. Tracked via `contractor_staff.staff_role_tier = MAIN_HANDLER`.
9. **Optionally track contractor roster** — for contractors who want it; same `contractor_staff` table with `staff_role_tier = OPTIONAL_ROSTER`. Most contractors leave this empty; the operator system doesn't require it.

What this module does NOT do

- **Doesn't own tech region definitions.** Region IDs, boundaries, candidate group codes — all ILM-CFG-02. EM-02 carries `tech_region_id` references only.
- **Doesn't define service scopes.** The set of valid service scope values (INSTALL, SUPPORT, MAINTENANCE, RECOVERY, AUDIT) is operator-extensible and lives in a separate catalog or operator config. EM-02 enforces no canonical list at the schema level.
- **Doesn't pick the technician for a specific WO.** EM-02 picks the contractor + commits to a slot. Which individual staff member from that contractor goes to that WO is the contractor's internal scheduling — not exposed to the operator system. (Some operators may surface this later via a staff-assignment-on-WO field; out of v1.0 for EM-02.)
- **Doesn't require a full contractor staff roster.** OPTIONAL_ROSTER staff records are opt-in per contractor. The operator-contractor relationship runs through the main handler; the rest is internal to the contractor.
- **Doesn't track staff real-time location.** Where a tech currently is, what they're doing — not EM-02. WO module tracks WO state; nothing tracks tech location in v1.0.
- **Doesn't filter WO routing by individual staff skills.** Contractor-level certification (`contractor_region_skill`) is the routing filter. Per-staff skill data isn't tracked at all.

- **Doesn't track per-staff skill certifications.** BSS is not an HR or training system. Operator HR / training departments already run systems that track who is qualified for what; replicating that in BSS adds maintenance burden with no operational payoff (WO routing uses contractor-level cert, not per-person).
- **Doesn't own staff personal data** beyond identity + role. Payroll, leave, performance metrics, personal contact details — out of EM-02 v1.0 scope. Future HRM module owns those.
- **Doesn't manage contractor contracts as legal documents.** Contract reference field is just a string identifier; the actual contract document, terms, vendor SLAs — live in an external contract management system or future procurement DD.
- **Doesn't decide work order priority within a slot.** If a slot has 3 capacity remaining and 5 WOs come in, EM-02's commitment API returns "INSUFFICIENT_CAPACITY" for the overflow; the operator's WO module decides queueing / rescheduling policy.

Regional flexibility

Contractor structures vary significantly between operators:

- **WIK (Kenya)** — hybrid: large internal team (wik-internal-tech-team) plus 6-8 franchise contractors across Nairobi neighborhoods + Mombasa + Diani. Franchises are small (10-20 staff each); internal team is larger. Mix of full-time staff + commission agents.
- **WUG (Uganda)** — primarily franchise model; 3-4 contractors across Kampala divisions. Smaller scale, lighter administrative overhead.
- **WTZ (Tanzania)** — TBD pending workshop; assumed smaller franchise network.
- **YASSN (Senegal)** — Dakar-centric; mix of franchise and direct-employee model; bilingual French + Wolof staff records.

The schema accommodates:

- Multiple contractor types (FRANCHISE / INTERNAL / SUB_CONTRACTOR) — operator-extensible
- Multi-region contractors (same contractor with rows in multiple contractor_region_scope)
- Multi-skill contractors with operator-extensible skill catalogs
- Multiple availability slots per (contractor, region, scope) for fine-grained scheduling
- Bilingual staff and contractor display names

Glossary

Term	Meaning
Contractor	A company providing field services. Three types: FRANCHISE (independent business under contract), INTERNAL (Wananchi-owned pseudo-contractor for internal teams — e.g., wik-internal-tech-team), or SUB_CONTRACTOR (sub-contracted by another contractor; multi-level).
Service scope	What kind of field work this contractor can do: INSTALL, SUPPORT, MAINTENANCE, RECOVERY, AUDIT. Operator-extensible. A contractor may have different scopes in different regions.
Region coverage	The (contractor, tech_region_id, service_scope) relationship with effective dates. The presence of an active row means "this contractor can do this scope in this region".
Skill	A technical competency code (fiber-install, ftth-cpe, vip-handling, ...). Operator-extensible. Required for some WO routing decisions.
Contractor skill	The relationship (contractor, region, skill) — "this contractor is certified to perform this skill in this region". Region-specific because regulatory requirements may differ.
Availability slot	A time-window with concurrency cap: (contractor, region, service_scope, day_of_week, hour_range, max_concurrent). The slot is the unit against which capacity is reserved for WOs.
Slot commitment	An immutable record: "this WO consumed N units of capacity from this slot at this time". The slot's remaining capacity = max_concurrent - count(active commitments).
Staff	An individual person associated with a contractor. Has FOUNDATION_AUTH user ID, primary role (installer / support_tech / team_leader / dispatcher / agent / main_handler), active status, and a staff role tier (MAIN_HANDLER or OPTIONAL_ROSTER).

Term	Meaning
Staff role tier	Either MAIN_HANDLER (mandatory; exactly one per active contractor; the operator's spoke for all communication with that contractor) or OPTIONAL_ROSTER (opt-in; additional staff for contractors who want full roster visibility).
Main handler	The one staff member per contractor who is the operator's official point of contact for assignments, slot scheduling, dispatching, escalation, and acknowledgments. ICN-01 routes contractor-facing notifications to the main handler's FOUNDATION_AUTH user.

2. Tables overview + relationships

2.1 Tables overview

EM-02 owns seven tables:

Table	Purpose
contractor	The contractor company / internal team / sub-contractor master record.
contractor_region_scope	Per-region service-scope coverage with effective dates. Time-versioned.
skill_catalog	Operator-extensible catalog of technical skill codes.
contractor_region_skill	Which skills a contractor is certified for in each region.
contractor_availability_slot	Day-of-week + hour-range windows with concurrency caps.
contractor_slot_commitment	Immutable ledger of WO commitments against slots. Capacity is derived.
contractor_staff	Main handler (mandatory, one per contractor) + optional roster, per contractor.

Full DDL in §3.

2.2 Entity relationships

[Missing image: file:///home/claude/diagrams/em2_erd.png]

Reading the diagram:

- **contractor** is the root. Four children hang off it: region-scope coverage, region-skill profile, availability slots, staff.
- **tech_region** (ILM-CFG-02) is referenced by region-scope, region-skill, and availability-slot rows; EM-02 holds the FK, ILM-CFG-02 owns the entity.
- **skill_catalog** is referenced by **contractor_region_skill** (contractor-level certification). This is the only skill linkage; individual staff are not tied to skills.
- **contractor_availability_slot** defines capacity windows; **contractor_slot_commitment** records consumption per WO. The slot's remaining capacity = `max_concurrent - count(active commitments)`.
- **work_order** (WO module) is the source of slot commitments; EM-02 carries the `wo_id` reference but doesn't own the WO entity.
- **contractor_staff** is a flat list of contractor contacts (one mandatory MAIN_HANDLER + optional OPTIONAL_ROSTER entries). No skill linkage from this table — BSS does not track per-person qualifications.

2.3 Cross-DD anchoring

- **WO module Drools assign-contractor rule** — calls `POST /api/contractor-availability` (the hot path). Receives ranked contractor list with slot IDs; commits to a slot via `POST /api/contractor-slot-commitments`. On WO completion or cancellation, releases the commitment.
- **ILM-CFG-02** — provides `tech_region_id` entities that EM-02 references in **contractor_region_scope**, **contractor_region_skill**, **contractor_availability_slot**. One-way FK from EM-02 to ILM-CFG-02; no reverse.
- **OSR-01** — retires `osr_contractor_stub` by pointing `stock_location.contractor_id` at EM-02's `contractor.contractor_id`. Retirement plan in §7.
- **ICN-01** — when a notification fires for a candidate group (e.g., `tech-region-NRB-LAVINGTON`), ICN-01 resolves group membership via `FOUNDATION_AUTH`. The `contractor_staff.foundation_auth_user_id` is what makes a staff

member discoverable. EM-02 maintains this linkage; ICN-01 doesn't query EM-02 directly.

- **FOUNDATION_AUTH** — staff foundation_auth_user_id references FOUNDATION_AUTH users; EM-02 carries the ID, FOUNDATION_AUTH owns identity + group memberships.

3. Schemas

3.1 contractor

```
CREATE TABLE contractor (
  contractor_id      TEXT PRIMARY KEY,
  operator_code      TEXT NOT NULL,
  contractor_type    TEXT NOT NULL,
  display_name_primary TEXT NOT NULL,
  display_name_secondary TEXT NULL,
  legal_entity_name  TEXT NULL,
  registration_number TEXT NULL,
  contract_reference TEXT NULL,
  primary_contact_name TEXT NULL,
  primary_contact_phone TEXT NULL,
  primary_contact_email TEXT NULL,
  active             BOOLEAN NOT NULL DEFAULT TRUE,
  effective_from     DATE NOT NULL,
  effective_to       DATE NULL,
  notes              TEXT NULL,
  created_at         TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at         TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_by         TEXT NOT NULL
);

-- 'contractor_ke_nairobi_a' | 'wik-internal-tech-team' | 'contractor_ke_nairobi_b'
-- 'FRANCHISE' | 'INTERNAL' | 'SUB_CONTRACTOR'
-- 'Franchise Nairobi A – Lavington Network Services Ltd'
-- bilingual; YASSN uses French
-- official registered name; same as display for FRANCHISE; NULL for INTERNAL
-- business registration / tax number; NULL for INTERNAL
-- external contract management system reference

CREATE INDEX idx_contractor_operator ON contractor (operator_code, active);
CREATE INDEX idx_contractor_type ON contractor (operator_code, contractor_type, active);
```

Sample data:

contractor_id	operator_code	contractor_type	display_name_primary	legal_entity_name	active	effective_from
wik-internal-tech-team	WIK	INTERNAL	WIK Internal Tech Team	NULL	true	2018-01-01
contractor_ke_nairobi_a	WIK	FRANCHISE	Franchise A – Lavington Network Services	Lavington Network Services Ltd	true	2022-04-01
contractor_ke_nairobi_b	WIK	FRANCHISE	Franchise B – backup Lavington	Backup Net Solutions Ltd	true	2022-04-01
contractor_ke_nairobi_e	WIK	FRANCHISE	Franchise E – Eastlands	Eastlands Fiber Co Ltd	true	2020-09-01
contractor_ke_nairobi_k	WIK	FRANCHISE	Franchise K – Kibera + Mukuru	Kibera Tech Services Ltd	true	2023-02-01
contractor_ke_mombasa_1	WIK	FRANCHISE	Former Mombasa Franchise (retired)	Mombasa Fiber Ltd (closed)	false	2019-06-01
contractor_ke_mombasa_2	WIK	FRANCHISE	Mombasa Franchise (current)	Coast Network Ltd	true	2025-01-01
contractor_sn_dakar_1	YASSN	FRANCHISE	Franchise principale Dakar centre	Dakar Réseaux SARL	true	2019-09-01
contractor_ug_kampala_central	WUG	FRANCHISE	Franchise Kampala Central Division	Central Kampala Networks Ltd	true	2022-08-01

Row 1 wik-internal-tech-team matches the ILM-CFG-02 reference pattern (internal pseudo-contractor for Karen / VIP customers). NULL legal_entity_name + registration_number since it's not a separate legal entity.

Row 6 contractor_ke_mombasa_1 has active=false matching ILM-CFG-02's contractor coverage termination at 2024-12-31 (corresponding effective_to on its coverage rows in §3.2 below).

3.2 contractor_region_scope

Per-region service-scope coverage. A contractor doing both INSTALL and SUPPORT in a region has TWO rows.

```
CREATE TABLE contractor_region_scope (
  coverage_id          TEXT PRIMARY KEY,          -- 'cov_em_01HZ...'
  contractor_id        TEXT NOT NULL REFERENCES contractor(contractor_id),
  tech_region_id       TEXT NOT NULL,             -- FK to ILM-CFG-02 tech_region; not enforced cross-module
  service_scope        TEXT NOT NULL,             -- 'INSTALL' | 'SUPPORT' | 'MAINTENANCE' | 'RECOVERY' | 'AUDIT'
  coverage_role        TEXT NOT NULL,             -- 'PRIMARY' | 'BACKUP' | 'EXCLUSIVE'
  effective_from       DATE NOT NULL,
  effective_to         DATE NULL,
  notes               TEXT NULL,
  created_at          TIMESTAMP NOT NULL DEFAULT NOW(),
  created_by          TEXT NOT NULL
);

CREATE INDEX idx_crs_contractor ON contractor_region_scope (contractor_id, effective_to);
CREATE INDEX idx_crs_region    ON contractor_region_scope (tech_region_id, service_scope, effective_to);
CREATE INDEX idx_crs_active    ON contractor_region_scope (tech_region_id, service_scope) WHERE effective_to IS NULL;
```

coverage_role:

- PRIMARY — first-choice contractor for this (region, scope). The WO module's Drools rule prefers PRIMARY.
- BACKUP — fallback when PRIMARY has no capacity. Multiple BACKUPS allowed; ordered by something other than this column (e.g., performance score from EM-04 Reporting).
- EXCLUSIVE — only contractor permitted for this (region, scope); no BACKUP allowed. Used for high-security or specialized scopes.

Sample data:

coverage_id	contractor_id	tech_region_id	service_scope	coverage_role	effective_from	effective_to
cov_em_01HZ_KAREN_INTERNAL_INSTALL	wik-internal-tech-team	KE-NRB-KAREN	INSTALL	PRIMARY	2018-01-01	NULL
cov_em_01HZ_KAREN_INTERNAL_SUPPORT	wik-internal-tech-team	KE-NRB-KAREN	SUPPORT	PRIMARY	2018-01-01	NULL
cov_em_01HZ_LAV_A_INSTALL	contractor_ke_nairobi_a	KE-NRB-LAVINGTON	INSTALL	PRIMARY	2022-04-01	NULL
cov_em_01HZ_LAV_A_SUPPORT	contractor_ke_nairobi_a	KE-NRB-LAVINGTON	SUPPORT	PRIMARY	2022-04-01	NULL
cov_em_01HZ_LAV_B_INSTALL	contractor_ke_nairobi_b	KE-NRB-LAVINGTON	INSTALL	BACKUP	2022-04-01	NULL
cov_em_01HZ_LAV_B_SUPPORT	contractor_ke_nairobi_b	KE-NRB-LAVINGTON	SUPPORT	BACKUP	2022-04-01	NULL
cov_em_01HZ_EAST_E_INSTALL	contractor_ke_nairobi_e	KE-NRB-EASTLANDS	INSTALL	PRIMARY	2020-09-01	NULL
cov_em_01HZ_KIB_K_INSTALL	contractor_ke_nairobi_k	KE-NRB-KIBERA	INSTALL	PRIMARY	2023-02-01	NULL
cov_em_01HZ_NYALI_M1_INSTALL	contractor_ke_mombasa_1	KE-MOMBASA-NYALI	INSTALL	PRIMARY	2019-06-01	2024-12-31
cov_em_01HZ_NYALI_M2_INSTALL	contractor_ke_mombasa_2	KE-MOMBASA-NYALI	INSTALL	PRIMARY	2025-01-01	NULL
cov_em_01HZ_DIANI_M2_INSTALL	contractor_ke_mombasa_2	KE-DIANI	INSTALL	PRIMARY	2025-01-01	NULL
cov_em_01HZ_PLATEAU_DKR1_INSTALL	contractor_sn_dakar_1	SN-DKR-PLATEAU	INSTALL	PRIMARY	2019-09-01	NULL

Row 9 + 10 show the Mombasa contractor handover: M1 PRIMARY until 2024-12-31, M2 PRIMARY from 2025-01-01. Historical rows are preserved.

Row 11 — contractor_ke_mombasa_2 covers both KE-MOMBASA-NYALI AND KE-DIANI as PRIMARY for INSTALL (matches ILM-CFG-02 candidate group sharing pattern).

3.3 skill_catalog

Operator-extensible catalog of technical skill codes.

```
CREATE TABLE skill_catalog (
  operator_code        TEXT NOT NULL,
```

```

skill_code          TEXT NOT NULL,                -- 'fiber-install' | 'ftth-cpe' | 'dwdm-config' | 'vip-handling'
display_name_primary TEXT NOT NULL,
display_name_secondary TEXT NULL,
description         TEXT NULL,
category           TEXT NOT NULL,                -- 'TECHNICAL_INSTALL' | 'TECHNICAL_SUPPORT' | 'SOFT_SKILL' | 'COMPLIANCE'
active             BOOLEAN NOT NULL DEFAULT TRUE,
created_at         TIMESTAMP NOT NULL DEFAULT NOW(),
updated_at         TIMESTAMP NOT NULL DEFAULT NOW(),
updated_by         TEXT NOT NULL,
PRIMARY KEY (operator_code, skill_code)
);

CREATE INDEX idx_skill_active   ON skill_catalog (operator_code, active);
CREATE INDEX idx_skill_category ON skill_catalog (operator_code, category, active);

```

Sample data:

operator_code	skill_code	display_name_primary	category	active
WIK	fiber-install	Fiber-optic installation	TECHNICAL_INSTALL	true
WIK	ftth-cpe	FTTH CPE configuration	TECHNICAL_INSTALL	true
WIK	dwdm-config	DWDM configuration	TECHNICAL_INSTALL	true
WIK	splicing	Optical splicing	TECHNICAL_INSTALL	true
WIK	vip-handling	VIP customer handling	SOFT_SKILL	true
WIK	iptv-stb-setup	IPTV STB setup + provisioning	TECHNICAL_INSTALL	true
WIK	network-troubleshooting	Network troubleshooting + fault diagnosis	TECHNICAL_SUPPORT	true
WIK	vehicle-license-class-b	Vehicle license – Class B (light commercial)	COMPLIANCE	true
YASSN	fiber-install	Installation fibre optique	TECHNICAL_INSTALL	true
YASSN	ftth-cpe	Configuration CPE FTTH	TECHNICAL_INSTALL	true

Skills are operator-scoped: a WIK skill_code is independent of a YASSN skill_code of the same name. Operator chooses the catalog.

3.4 contractor_region_skill

Which skills a contractor is certified to perform in each region.

```

CREATE TABLE contractor_region_skill (
  contractor_id      TEXT NOT NULL REFERENCES contractor(contractor_id),
  tech_region_id     TEXT NOT NULL,                -- FK to ILM-CFG-02 tech_region
  operator_code      TEXT NOT NULL,                -- denormalized for FK to skill_catalog
  skill_code         TEXT NOT NULL,
  certified_at       DATE NOT NULL,                -- when the contractor was certified for this skill in this region
  certified_by       TEXT NULL,                    -- 'wik-internal-training' | 'huawei-academy' | etc.
  active            BOOLEAN NOT NULL DEFAULT TRUE,
  notes             TEXT NULL,
  created_at        TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at        TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_by        TEXT NOT NULL,
  PRIMARY KEY (contractor_id, tech_region_id, skill_code),
  FOREIGN KEY (operator_code, skill_code) REFERENCES skill_catalog(operator_code, skill_code)
);

CREATE INDEX idx_crsk_region ON contractor_region_skill (tech_region_id, skill_code) WHERE active = TRUE;
CREATE INDEX idx_crsk_skill  ON contractor_region_skill (operator_code, skill_code);

```

Sample data:

contractor_id	tech_region_id	skill_code	certified_at	certified_by	active
wik-internal-tech-team	KE-NRB-KAREN	fiber-install	2018-01-01	wik-internal-training	true
wik-internal-tech-team	KE-NRB-KAREN	ftth-cpe	2018-01-01	huawei-academy	true
wik-internal-tech-team	KE-NRB-KAREN	vip-handling	2019-03-15	wik-customer-service-training	true

contractor_id	tech_region_id	skill_code	certified_at	certified_by	active
wik-internal-tech-team	KE-NRB-KAREN	iptv-stb-setup	2020-06-01	wik-internal-training	true
wik-internal-tech-team	KE-NRB-KAREN	network-troubleshooting	2018-01-01	wik-internal-training	true
contractor_ke_nairobi_a	KE-NRB-LAVINGTON	fiber-install	2022-04-01	wik-franchise-onboarding	true
contractor_ke_nairobi_a	KE-NRB-LAVINGTON	ftth-cpe	2022-04-01	wik-franchise-onboarding	true
contractor_ke_nairobi_a	KE-NRB-LAVINGTON	network-troubleshooting	2022-04-01	wik-franchise-onboarding	true
contractor_ke_nairobi_e	KE-NRB-EASTLANDS	fiber-install	2020-09-01	wik-franchise-onboarding	true
contractor_ke_nairobi_e	KE-NRB-EASTLANDS	ftth-cpe	2020-09-01	wik-franchise-onboarding	true
contractor_ke_mombasa_2	KE-MOMBASA-NYALI	fiber-install	2025-01-01	wik-franchise-onboarding	true
contractor_ke_mombasa_2	KE-MOMBASA-NYALI	ftth-cpe	2025-01-01	wik-franchise-onboarding	true
contractor_ke_mombasa_2	KE-DIANI	fiber-install	2025-01-01	wik-franchise-onboarding	true
contractor_ke_mombasa_2	KE-DIANI	ftth-cpe	2025-01-01	wik-franchise-onboarding	true
contractor_sn_dakar_1	SN-DKR-PLATEAU	fiber-install	2019-09-01	yassn-training	true
contractor_sn_dakar_1	SN-DKR-PLATEAU	ftth-cpe	2019-09-01	yassn-training	true

Internal team has VIP handling certification (matches the Hon. Wanjiku scenario from ILM-CFG-02 §7.2: VIP customers routed to internal team partly because they hold vip-handling). Franchise A does NOT have vip-handling; the WO Drools rule excludes them for VIP scopes.

Mombasa Franchise 2 holds the same skill set in BOTH Mombasa-Nyali AND Diani — supports the consolidated coverage pattern.

3.5 contractor_availability_slot

Day-of-week + hour-range windows with concurrency caps. The unit against which WO capacity is reserved.

```

CREATE TABLE contractor_availability_slot (
  slot_id          TEXT PRIMARY KEY,          -- 'slot_em_01HZ...'
  contractor_id    TEXT NOT NULL REFERENCES contractor(contractor_id),
  tech_region_id   TEXT NOT NULL,             -- FK to ILM-CFG-02 tech_region
  service_scope    TEXT NOT NULL,             -- INSTALL | SUPPORT | MAINTENANCE | etc.
  day_of_week      TEXT NOT NULL,             -- 'MONDAY' | 'TUESDAY' | ... | 'SUNDAY' | 'ALL_WEEK'
  hour_start       TIME NOT NULL,              -- e.g., '09:00'
  hour_end         TIME NOT NULL,              -- e.g., '18:00'
  timezone         TEXT NOT NULL,              -- 'Africa/Nairobi' | 'Africa/Dakar' | etc.
  max_concurrent   INTEGER NOT NULL,           -- capacity cap; e.g., 5 = up to 5 concurrent WOs
  emergency_only   BOOLEAN NOT NULL DEFAULT FALSE, -- TRUE = only used for high-priority WOs (e.g., overnight emergency)
  active          BOOLEAN NOT NULL DEFAULT TRUE,
  effective_from   DATE NOT NULL,
  effective_to     DATE NULL,
  notes           TEXT NULL,
  created_at       TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_at       TIMESTAMP NOT NULL DEFAULT NOW(),
  updated_by      TEXT NOT NULL,
  CHECK (max_concurrent > 0),
  CHECK (hour_start < hour_end)
);

CREATE INDEX idx_slot_contractor ON contractor_availability_slot (contractor_id, active);
CREATE INDEX idx_slot_region    ON contractor_availability_slot (tech_region_id, service_scope, active);
CREATE INDEX idx_slot_day       ON contractor_availability_slot (tech_region_id, service_scope, day_of_week, active);

```

Sample data:

slot_id	contractor_id	tech_region_id	service_scope	day_of_week	hour_start	hour_end	timezone	max_concurrent	emergency_only
slot_em_01 HZ_INT_KAREN_INSTALL_MON	wik-internal-tech-team	KE-NRB-KAREN	INSTALL	MONDAY	09:00	18:00	Africa/Nairobi	5	false
slot_em_01 HZ_INT_KAREN_INSTALL_TUE	wik-internal-tech-team	KE-NRB-KAREN	INSTALL	TUESDAY	09:00	18:00	Africa/Nairobi	5	false
slot_em_01 HZ_INT_KAREN_INSTALL_WED	wik-internal-tech-team	KE-NRB-KAREN	INSTALL	WEDNESDAY	09:00	18:00	Africa/Nairobi	5	false
slot_em_01 HZ_INT_KAREN_INSTALL_THU	wik-internal-tech-team	KE-NRB-KAREN	INSTALL	THURSDAY	09:00	18:00	Africa/Nairobi	5	false
slot_em_01 HZ_INT_KAREN_INSTALL_FRI	wik-internal-tech-team	KE-NRB-KAREN	INSTALL	FRIDAY	09:00	13:00	Africa/Nairobi	2	false
slot_em_01 HZ_INT_KAREN_SUP_ALL_WEEK	wik-internal-tech-team	KE-NRB-KAREN	SUPPORT	ALL_WEEK	08:00	20:00	Africa/Nairobi	3	false
slot_em_01 HZ_INT_KAREN_SUP_EMERG	wik-internal-tech-team	KE-NRB-KAREN	SUPPORT	ALL_WEEK	20:00	08:00	Africa/Nairobi	1	true
slot_em_01 HZ_LAV_A_INSTALL_MON	contractor_ke_nairobi_a	KE-NRB-LAVINGTON	INSTALL	MONDAY	08:00	19:00	Africa/Nairobi	8	false
slot_em_01 HZ_LAV_A_INSTALL_TUE	contractor_ke_nairobi_a	KE-NRB-LAVINGTON	INSTALL	TUESDAY	08:00	19:00	Africa/Nairobi	8	false
slot_em_01 HZ_LAV_A_INSTALL_SAT	contractor_ke_nairobi_a	KE-NRB-LAVINGTON	INSTALL	SATURDAY	09:00	15:00	Africa/Nairobi	3	false
slot_em_01 HZ_NYALI_M2_INSTALL_MON	contractor_ke_mombasa_2	KE-MOMBASA-NYALI	INSTALL	MONDAY	08:30	17:30	Africa/Nairobi	4	false
slot_em_01 HZ_DIANI_M2_INSTALL_MON	contractor_ke_mombasa_2	KE-DIANI	INSTALL	MONDAY	09:00	17:00	Africa/Nairobi	2	false
slot_em_01 HZ_PLATEAU_DKR1_INSTALL_MON	contractor_sn_dakar_1	SN-DKR-PLATEAU	INSTALL	MONDAY	08:30	18:00	Africa/Dakar	6	false

Rows 1-5 show the internal team's INSTALL schedule for KE-NRB-KAREN: 5-day work week, Mon-Thu full days at capacity 5; Friday half-day at capacity 2 (e.g., end-of-week wind-down or admin time).

Row 6 + 7 show SUPPORT split between regular hours (cap 3, all week 08:00-20:00) and emergency-only overnight (cap 1, 20:00-08:00). The emergency_only=true flag means only high-priority/critical WOs can consume this slot.

Row 10 — Franchise A has Saturday slots (cap 3); the internal team doesn't (no row for Saturday in their data).

The day_of_week= 'ALL_WEEK' shorthand on row 6 + 7 means "every day". Drools rule must evaluate slot eligibility against the WO's requested datetime.

3.6 contractor_slot_commitment

Immutable ledger: every WO committed to a slot is one row. Status moves ACTIVE → CONSUMED (WO completed) → RELEASED (cancelled) or EXPIRED (slot window passed without consumption).

```
CREATE TABLE contractor_slot_commitment (
  commitment_id      TEXT PRIMARY KEY,          -- 'cmt_01HZ...'
  slot_id             TEXT NOT NULL REFERENCES contractor_availability_slot(slot_id),
```

```

    contractor_id      TEXT NOT NULL REFERENCES contractor(contractor_id), -- denormalized for queries
    wo_id              TEXT NOT NULL, -- WO module reference
    committed_for_datetime TIMESTAMP NOT NULL, -- the actual scheduled WO datetime
    qty                INTEGER NOT NULL DEFAULT 1, -- almost always 1; allows multi-unit demand if needed
    status              TEXT NOT NULL, -- 'ACTIVE' | 'CONSUMED' | 'RELEASED' | 'EXPIRED'
    consumed_at         TIMESTAMP NULL, -- when WO completed (status → CONSUMED)
    released_at         TIMESTAMP NULL, -- when WO cancelled (status → RELEASED)
    expired_at          TIMESTAMP NULL, -- when slot window passed without consumption
    notes               TEXT NULL,
    created_at          TIMESTAMP NOT NULL DEFAULT NOW(),
    created_by          TEXT NOT NULL,
    CHECK (qty > 0),
    CHECK ((status = 'CONSUMED' AND consumed_at IS NOT NULL) OR
           (status = 'RELEASED' AND released_at IS NOT NULL) OR
           (status = 'EXPIRED' AND expired_at IS NOT NULL) OR
           (status = 'ACTIVE' AND consumed_at IS NULL AND released_at IS NULL AND expired_at IS NULL))
);

CREATE INDEX idx_commitment_slot_active ON contractor_slot_commitment (slot_id, committed_for_datetime) WHERE status = 'ACTIVE';
CREATE INDEX idx_commitment_wo ON contractor_slot_commitment (wo_id);
CREATE INDEX idx_commitment_contractor ON contractor_slot_commitment (contractor_id, committed_for_datetime DESC);

```

Sample data:

commitment_id	slot_id	contractor_id	wo_id	committed_for_datetime	qty	status
cmt_01HZ_WANJIKU_KAREN	slot_em_01HZ_INSTALL_MON	wik-internal-tech-team	wo_01HZ_WANJIKU_INSTALL	2026-02-23 14:00:00	1	CONSUMED
cmt_01HZ_LAV_BATCH_001_A	slot_em_01HZ_LAV_A_INSTALL_MON	contractor_ke_nairobi_a	wo_01HZ_LAV_001	2026-05-25 10:00:00	1	ACTIVE
cmt_01HZ_LAV_BATCH_001_B	slot_em_01HZ_LAV_A_INSTALL_MON	contractor_ke_nairobi_a	wo_01HZ_LAV_002	2026-05-25 14:00:00	1	ACTIVE
cmt_01HZ_LAV_BATCH_001_C	slot_em_01HZ_LAV_A_INSTALL_MON	contractor_ke_nairobi_a	wo_01HZ_LAV_003	2026-05-25 11:00:00	1	ACTIVE
cmt_01HZ_NYALI_M2_001	slot_em_01HZ_NYALI_M2_INSTALL_MON	contractor_ke_mombasa_2	wo_01HZ_NYALI_001	2026-05-25 09:00:00	1	RELEASED
cmt_01HZ_PLATEAU_001	slot_em_01HZ_PLATEAU_DKR1_INSTALL_MON	contractor_sn_dakar_1	wo_01HZ_PLAT_001	2026-05-25 14:00:00	1	ACTIVE

Row 1 — Hon. Wanjiku's install consumed the internal team's Monday slot on 2026-02-23 (matches the ILM-CFG-02 walkthrough). status=CONSUMED since WO completed.

Rows 2-4 — Franchise A has 3 active commitments for next Monday (2026-05-25). With max_concurrent=8, remaining capacity is 5.

Row 5 — Mombasa-Nyali commitment that was RELEASED (WO cancelled before install). Slot capacity is restored.

The slot's remaining capacity is computed live: `max_concurrent - count(commitments with status='ACTIVE' and committed_for_datetime within slot window)`.

3.7 contractor_staff

Individual staff members per contractor. Two tiers:

- **MAIN_HANDLER** — mandatory, exactly one active per active contractor. The operator's spoke for all communication with that contractor. ICN-01 routes contractor-facing notifications to this person.
- **OPTIONAL_ROSTER** — opt-in roster of additional staff. Most contractors leave this empty.

```

CREATE TABLE contractor_staff (
    staff_id          TEXT PRIMARY KEY, -- 'staff_em_01HZ...' or operator-readable 'main_wik_a01'
    contractor_id     TEXT NOT NULL REFERENCES contractor(contractor_id),
    operator_code     TEXT NOT NULL,
    staff_role_tier   TEXT NOT NULL, -- 'MAIN_HANDLER' | 'OPTIONAL_ROSTER'
    foundation_auth_user_id TEXT NULL, -- linkage to FOUNDATION_AUTH; MUST be set for MAIN_HANDLER; NULL for others
    display_name_primary TEXT NOT NULL, -- 'Grace Wairimu'
    display_name_secondary TEXT NULL,
    primary_role      TEXT NOT NULL, -- 'main_handler' (for tier=MAIN_HANDLER) | 'installer' | 'supervisor'
    employment_type   TEXT NOT NULL, -- 'EMPLOYEE' | 'CONTRACT_AGENT' | 'COMMISSION_AGENT'
    contact_phone      TEXT NULL, -- direct phone; especially relevant for MAIN_HANDLER
    contact_email      TEXT NULL, -- direct email; especially relevant for MAIN_HANDLER
    start_date        DATE NOT NULL,
    end_date          DATE NULL,
);

```

```

    active                BOOLEAN NOT NULL DEFAULT TRUE,
    notes                 TEXT NULL,
    created_at            TIMESTAMP NOT NULL DEFAULT NOW(),
    updated_at            TIMESTAMP NOT NULL DEFAULT NOW(),
    updated_by            TEXT NOT NULL,
    CHECK (staff_role_tier IN ('MAIN_HANDLER', 'OPTIONAL_ROSTER')),
    CHECK (staff_role_tier = 'OPTIONAL_ROSTER' OR foundation_auth_user_id IS NOT NULL)
);

CREATE INDEX idx_staff_contractor ON contractor_staff (contractor_id, active);
CREATE INDEX idx_staff_auth_user ON contractor_staff (foundation_auth_user_id) WHERE foundation_auth_user_id IS NOT NULL;
CREATE INDEX idx_staff_role ON contractor_staff (operator_code, primary_role, active);
CREATE INDEX idx_staff_tier ON contractor_staff (contractor_id, staff_role_tier, active);

-- Enforce exactly one active MAIN_HANDLER per contractor
CREATE UNIQUE INDEX uq_staff_main_handler
ON contractor_staff (contractor_id)
WHERE staff_role_tier = 'MAIN_HANDLER' AND active = TRUE;

```

The partial UNIQUE index enforces R-EM-CS-13 (see §4): a contractor can have at most one active MAIN_HANDLER row. The application layer enforces the "at least one" side by rejecting contractor activation without a main handler.

Sample data:

staff_id	contractor_id	staff_role_tier	foundation_auth_user_id	display_name_primary	primary_role	active
main_wik_internal_01	wik-internal-tech-team	MAIN_HANDLER	bo_user_main_int_01	Sarah Njoroge	main_handler	true
main_wik_a01	contractor_ke_nairobi_a	MAIN_HANDLER	bo_user_main_a01	Grace Wairimu	main_handler	true
main_wik_e01	contractor_ke_nairobi_e	MAIN_HANDLER	bo_user_main_e01	Joseph Mutiso	main_handler	true
main_wik_k01	contractor_ke_nairobi_k	MAIN_HANDLER	bo_user_main_k01	Lydia Atieno	main_handler	true
main_wik_msa_02	contractor_ke_mombasa_2	MAIN_HANDLER	bo_user_main_msa02	Hassan Mwangi	main_handler	true
main_yassn_dkr1	contractor_sn_dakar_1	MAIN_HANDLER	bo_user_main_dkr1	Aissatou Sow	main_handler	true
tech_wik_k01	wik-internal-tech-team	OPTIONAL_ROSTER	bo_user_tech_k01	John Mwangi	installer	true
tech_wik_k02	wik-internal-tech-team	OPTIONAL_ROSTER	bo_user_tech_k02	Esther Wanjiru	installer	true
tech_wik_e01	contractor_ke_nairobi_e	OPTIONAL_ROSTER	bo_user_tech_e01	Samuel Kiprop	installer	true
tech_wik_e02	contractor_ke_nairobi_e	OPTIONAL_ROSTER	bo_user_tech_e02	Mary Achieng	installer	true
tech_wik_e03	contractor_ke_nairobi_e	OPTIONAL_ROSTER	bo_user_tech_e03	David Otieno	installer	false
agent_wik_door_05	contractor_ke_nairobi_a	OPTIONAL_ROSTER	bo_user_door_05	James Maina	agent	true

Reading the sample:

- Rows 1-6: every active contractor has exactly one active MAIN_HANDLER. Sarah Njoroge is the spoke for the internal team; Grace Wairimu for Franchise A; etc. These rows are mandatory.
- Rows 7-11: wik-internal-tech-team (internal) opts in to OPTIONAL_ROSTER and lists their installers — operationally useful for the internal team because Wananchi is the employer.
- Row 12: Franchise A opts in only enough to register one commission agent (door-to-door sales) — they don't list their installers.
- The other franchises (E, K, MSA-2, DKR-1) have NO OPTIONAL_ROSTER rows — only their main handler. This is the common pattern for external franchises.

WO routing in §5.1 (the hot path) does NOT query this table for skill filtering — it queries contractor_region_skill (contractor-level certification, §3.4). The contractor_staff table answers a different question: "who's the operator's contact at this contractor?" (MAIN_HANDLER) and optionally "who works there?" (OPTIONAL_ROSTER, where present).

4. Rules

Rule	Description		
R-EM-CS-1	A contractor with active=false is excluded from POST /api/contractor-availability results. In-flight slot commitments to deactivated contractors continue to honor existing WO completions; new commitments are rejected.		
R-EM-CS-2	coverage_role: a region+scope may have ONE EXCLUSIVE row (excludes all others), or one PRIMARY + zero-or-more BACKUP rows. A second PRIMARY row for the same (region, scope) is permitted but indicates load-share — Drools rule decides allocation.		
R-EM-CS-3	Skill certification at the contractor level (contractor_region_skill) is REQUIRED for the contractor to appear in POST /api/contractor-availability results filtered by required skills. Skills are a hard filter: a contractor lacking even one required skill is excluded. Per-staff skill data is not tracked by EM-02; the contractor-level certification is the only signal WO routing consults.		
R-EM-CS-4	Slot eligibility evaluation: for a WO with committed_for_datetime = X, a slot is eligible if (a) active=true, (b) committed_for_datetime falls within the slot's [hour_start, hour_end) window in the slot's timezone, (c) the WO's day-of-week matches day_of_week or day_of_week='ALL_WEEK', (d) emergency_only=false OR the WO carries an emergency flag, (e) effective_from <= today AND (effective_to IS NULL OR effective_to >= today).		
R-EM-CS-5	Slot capacity computation: remaining_capacity(slot, datetime) = max_concurrent - count(contractor_slot_commitment WHERE slot_id=slot AND status='ACTIVE' AND committed_for_datetime overlaps with the same operational window as datetime). The "operational window" for v1.0 is the same calendar day in the slot's timezone.		
R-EM-CS-6	Slot commitment atomicity: POST /api/contractor-slot-commitments either commits the row + decrements available capacity atomically, or returns INSUFFICIENT_CAPACITY with the current capacity state. No partial commits.		
R-EM-CS-7	Commitment lifecycle: ACTIVE → CONSUMED (WO completed, capacity is consumed for the period and not restored)	RELEASED (WO cancelled, capacity restored if the day hasn't passed)	EXPIRED (slot's day has passed without the WO completing — system flags for ops review). Status transitions are append-only via dedicated APIs.

Rule	Description		
R-EM-CS-9	Staff with active=false are excluded from FOUNDATION_AUTH group memberships (cleared by a daily reconciliation job). For MAIN_HANDLER staff specifically, deactivation requires a successor to be designated first (see R-EM-CS-13); the application layer enforces this before deactivation commits.		
R-EM-CS-10	Contractor + staff cross-operator scoping: every entity in EM-02 carries operator_code (directly or via FK chain to contractor.operator_code). Cross-operator queries require sophix-superadmin role.		
R-EM-CS-11	Availability slot timezone is captured per slot (not derived from operator) because cross-timezone operators may exist in future. v1.0 operators each have a single timezone; the column is forward-compatibility.		
R-EM-CS-12	A WO may consume capacity from multiple slots if its work spans multiple days (e.g., a multi-day install). Each day's commitment is a separate contractor_slot_commitment row sharing the same wo_id. v1.0 ships single-day commitments only; multi-day is supported by schema but uses one row per day.		
R-EM-CS-13	Main handler mandate. Every active contractor MUST have exactly one active contractor_staff row with staff_role_tier='MAIN_HANDLE R'. The partial UNIQUE index (uq_staff_main_handler) enforces "at most one"; the application layer enforces "at least one" by rejecting (a) contractor activation without a main handler, (b) main handler deactivation without successor designation. Main handler changes emit a ContractorMainHandlerChanged Kafka event for operator notification routing updates.		
R-EM-CS-14	contractor_staff.staff_role_tier='MAIN_HANDLER' rows MUST have a foundation_auth_user_id set (enforced by CHECK constraint). The main handler MUST be a real BO UI user since they receive ICN-01 notifications. OPTIONAL_ROSTER rows MAY have NULL foundation_auth_user_id (e.g., a back-office records-only entry for a non-login staff member).		

5. REST API

5.1 Contractor availability — the hot path

The endpoint WO module's Drools assign-contractor rule calls every time it needs to pick a contractor.

```
POST /api/contractor-availability HTTP/1.1
Authorization: Bearer <wo-service-jwt>
Content-Type: application/json
```

```
{
  "operatorCode":      "WIK",
  "techRegionId":     "KE-NRB-KAREN",
  "serviceScope":     "INSTALL",
```

```

    "requiredSkills": ["fiber-install", "ftth-cpe", "vip-handling"],
    "windowStart": "2026-02-23T14:00:00+03:00",
    "windowEnd": "2026-02-23T17:00:00+03:00",
    "concurrentDemand": 1,
    "preferEmergency": false
  }
}

→ 200 OK
{
  "operatorCode": "WIK",
  "techRegionId": "KE-NRB-KAREN",
  "evaluatedAt": "2026-02-17T10:00:00Z",
  "availableContractors": [
    {
      "contractorId": "wik-internal-tech-team",
      "contractorType": "INTERNAL",
      "coverageRole": "PRIMARY",
      "matchedSkills": ["fiber-install", "ftth-cpe", "vip-handling"],
      "slot": {
        "slotId": "slot_em_01HZ_INT_KAREN_INSTALL_MON",
        "dayOfWeek": "MONDAY",
        "hourStart": "09:00",
        "hourEnd": "18:00",
        "timezone": "Africa/Nairobi",
        "maxConcurrent": 5,
        "currentlyCommitted": 2,
        "remainingCapacity": 3
      },
      "rank": 1,
      "rankRationale": "INTERNAL + full skill match + PRIMARY coverage"
    }
  ],
  "totalAvailable": 1
}

```

If no contractor qualifies:

```

{
  "availableContractors": [],
  "totalAvailable": 0,
  "reason": "No contractor in KE-NRB-KAREN with required skills [vip-handling] has capacity in the requested window"
}

```

Filter logic (the Drools rule that WO module wraps):

1. Find all contractor_region_scope rows matching (tech_region_id, service_scope) with effective_to IS NULL.
2. Filter to contractors holding all requiredSkills per contractor_region_skill.
3. Filter to contractors with active=true.
4. For each, evaluate eligible slots per R-EM-CS-4 against windowStart/End.
5. For each eligible slot, compute remaining capacity per R-EM-CS-5.
6. Filter to slots with remaining_capacity >= concurrentDemand.
7. Rank by: coverage_role (PRIMARY > BACKUP), skill match completeness, remaining capacity ratio.
8. Return ranked list.

5.2 Slot commitment

```

POST /api/contractor-slot-commitments HTTP/1.1
Authorization: Bearer <wo-service-jwt>
Content-Type: application/json

```

```

{
  "slotId": "slot_em_01HZ_INT_KAREN_INSTALL_MON",
  "woId": "wo_01HZ_WANJIKU_INSTALL",
  "committedForDatetime": "2026-02-23T14:00:00+03:00",
  "qty": 1
}

```

→ 201 Created

```

{
  "commitmentId": "cmt_01HZ_WANJIKU_KAREN",
  "slotId": "slot_em_01HZ_INT_KAREN_INSTALL_MON",
  "remainingCapacityAfter": 2,
  "status": "ACTIVE"
}

```

Or rejected:

```
→ 409 Conflict
{
  "error": "INSUFFICIENT_CAPACITY",
  "slotId": "slot_em_01HZ_LAV_A_INSTALL_MON",
  "requestedQty": 1,
  "remainingCapacity": 0,
  "details": "Slot at capacity (8/8 committed) for 2026-05-25"
}

POST /api/contractor-slot-commitments/{id}/consume # transition to CONSUMED (WO completed)
DELETE /api/contractor-slot-commitments/{id} # transition to RELEASED (WO cancelled)
GET /api/contractor-slot-commitments?slotId=X&date=Y # query commitments per slot
GET /api/contractor-slot-commitments/{id} # detail
```

5.3 Contractor catalog

```
GET /api/contractors?operatorCode=WIK&active=true&contractorType=FRANCHISE
GET /api/contractors/{contractor_id} # detail
POST /api/contractors # create (BO admin only)
PATCH /api/contractors/{contractor_id} # update mutable fields
GET /api/contractors/{contractor_id}/region-scopes # list coverage rows
POST /api/contractors/{contractor_id}/region-scopes # add coverage
GET /api/contractors/{contractor_id}/region-skills # list certified skills
POST /api/contractors/{contractor_id}/region-skills # add skill certification
GET /api/contractors/{contractor_id}/availability-slots # list slots
POST /api/contractors/{contractor_id}/availability-slots # add slot
PATCH /api/contractor-availability-slots/{slot_id} # update slot (active, max_concurrent, etc.)
```

5.4 Skill catalog

```
GET /api/skill-catalog?operatorCode=WIK&active=true
POST /api/skill-catalog
PATCH /api/skill-catalog/{operator}/{skill_code}
```

5.5 Staff

```
GET /api/contractor-staff?contractorId=X&staffRoleTier=MAIN_HANDLER&active=true
GET /api/contractor-staff?contractorId=X&staffRoleTier=OPTIONAL_ROSTER&active=true
GET /api/contractor-staff?contractorId=X&active=true # both tiers
GET /api/contractor-staff/{staff_id}
POST /api/contractor-staff # create; tier required
PATCH /api/contractor-staff/{staff_id} # update mutable fields
```

Main-handler resolution — the common query for ICN-01 + WO module dispatch:

```
GET /api/contractors/{contractor_id}/main-handler HTTP/1.1
Authorization: Bearer <service-jwt>
```

```
→ 200 OK
{
  "contractorId": "contractor_ke_nairobi_a",
  "staffId": "main_wik_a01",
  "displayNamePrimary": "Grace Wairimu",
  "foundationAuthUserId": "bo_user_main_a01",
  "contactPhone": "+254712345678",
  "contactEmail": "grace@franchise-a.co.ke",
  "active": true
}
```

This is the hot path for "who at contractor X do I notify about this WO assignment?" — ICN-01 calls it after the slot commitment is made.

Main-handler change (handover) — requires successor designation; can't deactivate an active main handler without naming the next one:

```
POST /api/contractors/{contractor_id}/main-handler/handover HTTP/1.1
Authorization: Bearer <bo-admin-jwt>
Content-Type: application/json
```

```
{
  "outgoingStaffId": "main_wik_a01",
  "incomingStaff": {
    "displayNamePrimary": "Joseph Kibet",
    "primaryRole": "main_handler",
    "foundationAuthUserId": "bo_user_main_a02",
    "contactPhone": "+254712999888",
    "contactEmail": "joseph@franchise-a.co.ke",
    "employmentType": "EMPLOYEE",
    "startDate": "2026-06-01"
  }
}
```

```

    },
    "effectiveFrom":      "2026-06-01",
    "handoverNotes":      "Grace moving to support role; Joseph promoted to main handler"
  }
}

→ 201 Created
{
  "outgoingStaffId":      "main_wik_a01",
  "outgoingStatus":       "deactivated",
  "incomingStaffId":      "main_wik_a02",
  "incomingStatus":       "active",
  "handoverEventId":      "hndvr_01HZ_FRANCHISE_A_2026_06"
}

```

The outgoing main handler is automatically deactivated (and dropped from FOUNDATION_AUTH groups by the reconciliation job); the incoming is activated. A ContractorMainHandlerChanged Kafka event fires (see §6).

6. Kafka events

EM-02 publishes on `sophix.em.cs.*`.

Event type	Topic	Emitted when	Consumers
ContractorCreated	<code>sophix.em.cs.contractor-created</code>	New contractor row	OSR-01 (open van location), ICN-01 group setup
ContractorDeactivated	<code>sophix.em.cs.contractor-deactivated</code>	Contractor active=true → false	OSR-01 (close van location), WO module (reroute in-flight WOs)
ContractorRegionScopeChanged	<code>sophix.em.cs.region-scope-changed</code>	Coverage row added or effective_to set	WO module (Drools rule cache invalidation)
ContractorRegionSkillChanged	<code>sophix.em.cs.region-skill-changed</code>	Skill certification added/removed for contractor in region	WO module
ContractorAvailabilitySlotChanged	<code>sophix.em.cs.slot-changed</code>	Slot row created or modified	WO module (capacity recompute)
ContractorSlotCommitmentCreated	<code>sophix.em.cs.slot-commitment-created</code>	New commitment	EM-04 Reporting (capacity analytics)
ContractorSlotCommitmentConsumed	<code>sophix.em.cs.slot-commitment-consumed</code>	Commitment → CONSUMED	Ops dashboards, EM-04
ContractorSlotCommitmentReleased	<code>sophix.em.cs.slot-commitment-released</code>	Commitment → RELEASED	Ops dashboards
ContractorMainHandlerChanged	<code>sophix.em.cs.main-handler-changed</code>	Main handler handover applied (outgoing deactivated + incoming activated atomically)	WO module + ICN-01 (re-route active assignment comms); FOUNDATION_AUTH (provision/deprovision); ops dashboards
ContractorStaffCreated	<code>sophix.em.cs.staff-created</code>	New staff row	FOUNDATION_AUTH provisioning (if foundation_auth_user_id set), ICN-01 group membership setup (for MAIN_HANDLER). Event payload includes staffRoleTier.
ContractorStaffDeactivated	<code>sophix.em.cs.staff-deactivated</code>	Staff active=true → false	FOUNDATION_AUTH deprovisioning, ICN-01 group removal. For MAIN_HANDLER, this event fires only AFTER ContractorMainHandlerChanged (i.e., a successor has been designated). Event payload includes staffRoleTier.

6.1 Sample event — ContractorSlotCommitmentConsumed

```

{
  "eventType":      "ContractorSlotCommitmentConsumed",
  "aggregateId":    "cmt_01HZ_WANJIKU_KAREN",
  "timestamp":      "2026-02-23T15:45:00.000Z",
  "payload": {
    "operatorCode":      "WIK",
    "commitmentId":      "cmt_01HZ_WANJIKU_KAREN",
    "slotId":            "slot_em_01HZ_INT_KAREN_INSTALL_MON",
    "contractorId":      "wik-internal-tech-team",
    "techRegionId":      "KE-NRB-KAREN",
    "serviceScope":      "INSTALL",
  }
}

```

```

    "woId": "wo_01HZ_WANJIKU_INSTALL",
    "committedForDatetime": "2026-02-23T14:00:00+03:00",
    "consumedAt": "2026-02-23T15:45:00+03:00",
    "timeToConsumeMinutes": 105
  }
}

```

EM-04 Reporting (pending) consumes this for contractor performance metrics (mean time from commitment to completion).

6.2 Sample event — ContractorMainHandlerChanged

```

{
  "eventType": "ContractorMainHandlerChanged",
  "aggregateId": "contractor_ke_nairobi_a",
  "timestamp": "2026-06-01T08:00:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "contractorId": "contractor_ke_nairobi_a",
    "outgoingStaffId": "main_wik_a01",
    "outgoingDisplayName": "Grace Wairimu",
    "outgoingFoundationAuthUserId": "bo_user_main_a01",
    "incomingStaffId": "main_wik_a02",
    "incomingDisplayName": "Joseph Kibet",
    "incomingFoundationAuthUserId": "bo_user_main_a02",
    "incomingContactPhone": "+254712999888",
    "incomingContactEmail": "joseph@franchise-a.co.ke",
    "effectiveFrom": "2026-06-01T00:00:00+03:00",
    "handoverEventId": "hndvr_01HZ_FRANCHISE_A_2026_06",
    "handoverNotes": "Grace moving to support role; Joseph promoted to main handler",
    "actionRequired": "WO module + ICN-01 reroute active assignation comms to incoming main handler"
  }
}

```

WO module + ICN-01 are the critical consumers: any in-flight WO commitments that had been generating notifications to Grace's FOUNDATION_AUTH user must reroute to Joseph from this event onward.

7. Stub retirement

EM-02 retires `osr_contractor_stub` from OSR-01.

7.1 Retirement plan

The stub `osr_contractor_stub(contractor_id PK, contractor_name)` exists today in OSR-01 v1.0 to satisfy the FK reference from `stock_location.contractor_id`. EM-02's `contractor` table is the canonical replacement.

Migration steps:

1. **Pre-deployment:** EM-02 schema deployed alongside `osr_contractor_stub`. Both tables coexist.
2. **Backfill:** a one-time job iterates `osr_contractor_stub` rows. For each, insert a corresponding `contractor` row with `contractor_type='FRANCHISE'` (default; ops review for any INTERNAL ones), `effective_from=<reasonable historical date>`, and seed-script-derived display names. The `contractor_id` value is preserved verbatim (since `stock_location` rows already reference it).
3. **FK switch:** `stock_location.contractor_id` FK target updated from `osr_contractor_stub.contractor_id` to `contractor.contractor_id`. Since the `contractor_id` values are identical, this is a metadata change only — no row updates needed.
4. **Verification window:** 1 week with both tables coexisting. Spot-check that every active `stock_location` row resolves to a `contractor` row.
5. **Removal:** drop `osr_contractor_stub`. OSR-01 §3.1 / §10 documentation updated to reference EM-02 directly.

7.2 Caller-side changes

OSR-01 BO admin UI ("Contractors" tab) now reads from EM-02's `/api/contractors` endpoint. Previously it read from the stub; now full contractor metadata (legal entity, contract reference, contact details, skills, slots) is available.

WO module Drools rules previously had to fetch contractor info from EM-02 directly (since stub had only name); after retirement, the rules continue to call EM-02 — no code change required since stub-vs-EM-02 was already abstracted at the rule layer.

8. Cross-DD coordination

8.1 Upstream callers

Caller	Endpoint	When
WO module Drools assign-contractor rule	POST /api/contractor-availability	Every WO assignment (FLOW-INSTALLATION + FLOW-SHIFTING + FLOW-SUPPORT)
WO module FLOW-INSTALLATION	POST /api/contractor-slot-commitments	After contractor selection — reserve the slot capacity
WO module FLOW-* (completion)	POST /api/contractor-slot-commitments/{id}/consume	When WO finalises — slot consumed
WO module FLOW-* (cancellation)	DELETE /api/contractor-slot-commitments/{id}	WO cancelled — slot capacity released
OSR-01	GET /api/contractors?operatorCode=X	Contractor selection at BO admin time (open new van)
WO module FLOW-* (assignment comms)	GET /api/contractors/{id}/main-handler	After slot commitment — find the contractor's spoke person for assignment notification
ICN-01	indirect — via FOUNDATION_AUTH	Group membership lookup uses MAIN_HANDLER contractor_staff.foundation_auth_user_id + any opted-in OPTIONAL_ROSTER staff with login
BO admin UI	various	Contractor onboarding, slot scheduling, staff management

8.2 Downstream consumers (Kafka)

Consumer	Events subscribed	Action
WO module	ContractorRegionScopeChanged, ContractorRegionSkillChanged, ContractorAvailabilitySlotChanged	Invalidate routing rule caches
OSR-01	ContractorCreated, ContractorDeactivated	Open / close stock_location van rows
ICN-01	ContractorStaffCreated, ContractorStaffDeactivated, ContractorMainHandlerChanged	Update FOUNDATION_AUTH group memberships (via reconciliation job); refresh main-handler routing on handover
FOUNDATION_AUTH	ContractorStaffCreated, ContractorStaffDeactivated, ContractorMainHandlerChanged	Provision / deprovision user accounts
EM-04 Reporting (pending)	ContractorSlotCommitmentCreated, ContractorSlotCommitmentConsumed	Contractor performance metrics: WO completion rate, mean time to consume, slot utilization
BO admin ops dashboard	All sophix.em.cs.*	Contractor topology + performance display

8.3 Sibling modules

Module	Direction	Contract
ILM-CFG-02 (Tech Region Registry)	EM-02 → ILM-CFG-02 (read)	EM-02 references tech_region_id on contractor_region_scope, contractor_region_skill, contractor_availability_slot. ILM-CFG-02 is upstream; no reverse FK.
OSR-01 (Stock Chain)	OSR-01 → EM-02 (read, via stub retirement)	stock_location.contractor_id references contractor.contractor_id after retirement.
ICN-01 (Internal Communications)	ICN-01 ← EM-02 (via FOUNDATION_AUTH)	EM-02 maintains contractor_staff.foundation_auth_user_id for MAIN_HANDLER (mandatory) and OPTIONAL_ROSTER (where present); FOUNDATION_AUTH owns group membership; ICN-01 queries FOUNDATION_AUTH at dispatch time. Contractor-facing assignment notifications route to the contractor's MAIN_HANDLER; ContractorMainHandlerChanged events trigger ICN-01 to refresh group memberships.

Module	Direction	Contract
WO module	WO → EM-02 (heaviest reader + writer)	Drools rules call contractor-availability (filters by contractor_region_skill, NOT staff certs); slot commitments written + lifecycle-transitioned by WO module's actions. For assignation notifications WO module resolves the contractor's main handler via GET /api/contractors/{id}/main-handler.
FOUNDATION_AUTH	EM-02 ↔ FOUNDATION_AUTH	EM-02 carries the FK; FOUNDATION_AUTH owns identity + group memberships. A reconciliation job (daily) syncs contractor_staff active state to FOUNDATION_AUTH user enable/disable.
EM-04 Reporting (pending)	EM-04 ← EM-02 (via Kafka)	Performance metrics consumed from slot commitment events.

8.4 Dependencies (build order)

1. **ILM-CFG-02** (Tech Region Registry) — already shipped; provides tech_region_id references for region-scope + region-skill + slot rows.
2. **FOUNDATION_AUTH** — staff foundation_auth_user_id references require pre-existing FOUNDATION_AUTH users; provisioning workflow integrated via Kafka events.
3. **Skill catalog seed** — per-operator seed scripts populate skill_catalog with the operator's standard skill codes. WIK ships ~12 codes; YASSN ships ~10 codes (French names); WUG + WTZ smaller seeds.
4. **Contractor seed** — per-operator seed of existing contractors (migrated from osr_contractor_stub plus internal team rows) including their region scopes + skills + initial slot schedules.
5. **Staff seed** — per-operator staff data loaded from the operator's existing HR/contractor systems (out of scope for v1.0 schema — just data migration).
6. **OSR-01 stub retirement** — per §7; runs after EM-02 contractor seed is in place.
7. **WO module integration** — Drools assign-contractor rule deployed referencing EM-02's availability + commitment APIs.

9. Workshop sources

No dedicated workshop deep-dive for Contractor & Staff Registry. The functional area is implicit in:

- **Work Order workshop** (deep-dive 07) — contractor routing, dispatcher assignment, technician handover.
- **HPRFS workshop** (deep-dive 03) — homepass-to-contractor coverage relationships.
- **Customer Management workshop** (deep-dive 01) — references to franchise + internal team operational distinctions.

V3 design intent: **contractor-anchored capacity model with availability slots + concurrency caps** — the operator schedules against contractor slots, not against individual technicians; contractors internally manage their staff scheduling. Skill certifications track regulatory compliance + inform routing without coupling operator scheduling to individual staff. The TZ codebase's current handling of contractor + staff data is unverified pending cartography; placeholder: "TBD pending codebase review."

Out of v1.0 scope:

- **Per-staff WO assignment from operator side** — v1.0 commits to contractors + slots; the individual staff member assigned is the contractor's internal scheduling. Future enhancement may surface a staff_id on the WO row for visibility.
- **Real-time staff location tracking** — out.
- **Multi-day WO commitments as a single ledger row** — out; v1.0 uses one row per day.
- **Contractor performance scoring + auto-ranking** — out; EM-04 Reporting (pending) owns analytics.
- **Sub-contracting chain depth > 1** — v1.0 supports SUB_CONTRACTOR type but doesn't model sub-contracting hierarchies beyond simple typing.
- **HRM concerns** (payroll, leave, performance, personal contact details beyond identity) — out; future HRM module.
- **Multi-currency contractor billing** — out; contractor billing tracked outside V3.

- **Per-staff personal availability** (vacation, leave) — out; contractor manages this internally; reflected as reduced slot capacity if contractor chooses.

End of DD_EM-02-Contractor_and_Staff_Registry-v1.0.